Magnetic sensors

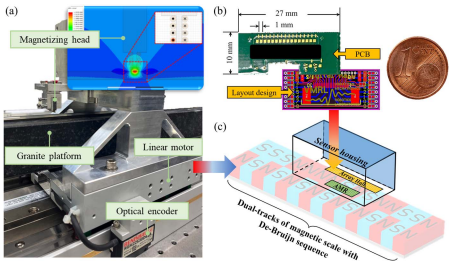
Generation of Absolute Patterns for Magnetic Encoders of Arbitrary Size With De Bruijn Sequences

Kai-Yang Peng[1] and Jen-Yuan Chang[2*]

[1]Department of Power Mechanical Engineering, National Tsing Hua University, Hsinchu 30013, Taiwan
[2]Department of Mechanical and Computer-Aided Engineering, National Formosa University, Yunlin 632, Taiwan
*Senior Member, IEEE

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Abstract—In this letter, we propose a magnetic encoding method utilizing De Bruijn sequences for absolute encoding. By integrating a single-dimensional array of Hall-effect sensors, a dual-track absolute positioning system was successfully developed and validated. The system leverages the Eulerian circuit algorithm to ensure the nonrepetitive nature of subsequences, fully exploiting variable symbols and subarray length characteristics within the sequence, thereby enabling arbitrary cyclic encoding lengths. This design overcomes the constraints imposed by commercially available products with fixed encoder dimensions. The feasibility of the system was validated through a specific magnetization process, resulting in a dual-track magnetic scale coupled with custom-made arrayed sensing elements and firmware signal integration. This design not only ensures high precision in positioning but also offers universality and adaptability of the generation of absolute patterns for quasi-absolute magnetic encoders of any size, making them applicable to diverse applications.



Index Terms—Magnetic sensors, De Bruijn (DB) sequence, dual-track, Hall-effect sensor, single-dimensional array.

I. INTRODUCTION

Magnetic position systems have been widely applied in various fields, including robotic arms, linear moving platforms, processing machines, and precision instruments. According to the magnetic encoding arrangement, it can be simply classified into two types: the incremental type and the absolute type, respectively. The incremental encoder has a magnetic medium with alternating arrangements of magnetic north and south poles [1]. The periodic signal outputted by the encoder has high reliability and high precision, which is suitable for identifying the relative movement distance between the sensor and the encoding medium. However, when the machine is restarted, the encoder cannot self-identify its current position but is required to return back to its origin so as to recalculate the relative movement distance. To solve this problem, special sequences can be created using different algorithms to define the absolute position on the magnetic medium. For example, a commonly used Vernier/Nonius code consists of two incremental tracks, where the absolute position is calculated by subtracting two periodic signals from the tracks, provided that the number of periods of the two signals differs by only one [2]. Considering commercial applications' confidentiality and patent restrictions, pseudorandom and De Bruijn (DB)-encoded designs are more prevalently found in commercial products. Pseudorandom encoding uses specific polynomial functions combined with XOR gates to generate seemingly random subsequences [3]. Similarly, DB encoding, which also appears random, is a cyclic sequence developed based on Eulerian paths, Eulerian circuits, and Hamiltonian cycles in graph theory [4]. From the perspective of the definition of absolute position, the encoding mechanism of vernier code is considered absolute; however, the latter designs are classified as quasi-absolute due to the initialization process involved.

Corresponding author: Jen-Yuan Chang (e-mail: jychang@pme.nthu.edu.tw).
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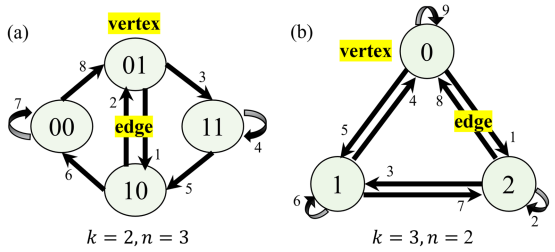


Fig. 1. Graphical illustrations of Eulerian circuits with different numbers of symbols: (a) B(2, 3) sequence consisting of symbols {0, 1}, and (b) B(3, 2) sequence is depicted, composed of symbols {0, 1, 2}.

Although numerous studies have been published on encoding, most of them focus on the impact of grating order on signals [5], [6] to evaluate system fault tolerance. In addition, different materials' conductive properties are utilized in conjunction with resonators to adapt to environmental needs [7], [8]. However, in the development of linear and rotary encoders, specific encoding formats limit the system's maximum stroke range. Even if partial encoding can meet various linear size requirements, in the case of rotary code discs, both radial and axial encoding directions encounter numerous limitations due to repetitive coding at the junctions. Furthermore, encoders are also restricted by the arrangement of sensing elements in commercially available sensing modules, thereby limiting product development.

Therefore, this study proposes the use of DB sequences and magnetic materials to address the limitations of absolute encoding in size development, demonstrating the excellent adaptability of DB sequences to arbitrary sizes. To enhance product confidentiality, the study employs a unique encoding format differing from binary to prevent reverse engineering, leveraging differences in magnetic pole polarity. In addition, to break free from the constraints of

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existing sensing modules, the experiment utilized specific arrangements of sensing elements, verifying that the custom quasi-absolute encoder has precise positioning capabilities. Although its resolution cannot compete with optical encoders, it is sufficient for most applications.

II. CYCLIC CODING METHOD

A. Principle of DB Sequence

A DB sequence, abbreviated as B(k, n), is a matrix composed of k distinct symbols, where all consecutive subsequences of length n precisely represent all permutations of k symbols with a length of k^n. Depending on application scenarios, the sorting pattern of the sequence can be classified into two categories, namely, the "Eulerian Path" and the "Eulerian circuit," respectively. The following points offer an explanation based on the sufficient and necessary conditions for the path of the sequence, as graphically depicted in Fig. 1.

1) Each edge on the graph must be connected.
2) To ensure that the sequence forms an Eulerian circuit, every vertex in the graph must have even degree, meaning that the vertex must be connected to an even number of other elements within the subarray with in-degree must be equal to its out-degree.

In the context of an Eulerian path, it differs from an Eulerian circuit in that an Eulerian path may commence and terminate at two distinct vertices, with the starting vertex having an out-degree exceeding its in-degree by one, that is out-degree minuses in-degree equals unity, while the ending vertex has an in-degree surpassing its out-degree by one, that is in-degree minuses out-degree equals one.

Here, consider the directed graph that corresponds to a simple DB sequence with an Eulerian circuit, as depicted in Fig. 1(a). For instance, the sequence 01011100, which is shorthand noted as DB B(23) sequence, has its consecutive subsequence looping from the end to the beginning, which 010, 101, 011, 111, 110, 100, 000, 001 precisely represent all sequences of length 3 composed of {0, 1}. When constructing the circuit graph, vertices are sorted from the last two digits of each subsequence, forming a cyclic encoding according to the order of edges from 1 to 8. Consequently, we can deduce that the DB sequence of B(k, n) possesses an Eulerian circuit, where k is the number of symbols used in the sequence and n being the number of order. For the case of $k = 3$ (utilizing symbols 0, 1, and 2), a DB circuit can similarly be depicted. The DB sequence B(32) as illustrated in Fig. 1(b) is 0221011200, and an equivalent Eulerian cycle is observed. In practical terms, the commonality between the two lies in each subsequence representing a specific position on a linear scale of length k^n, akin to multiple sets of indices marked on a scale. This aids in determining the absolute position of the test substance on the platform. The distinction arises when the number of symbols changes. Under the condition of a constant subsequence length, a greater number of symbols can generate longer cyclic sequences, and the number of DB sequences achievable through the combinations of parameters k and n [9], [10] can be expressed in (1). This implies that the effective stroke of the encoder can extend further or that a rotary encoder possesses a larger diameter.

$$N_{B(k,\ n)} = \frac{(k!)^{k^{n-1}}}{k^n};\ k > 1. \tag{1}$$

B. Sequence With Arbitrary Length

Referring to the comparison of commercially available magnetic encoding technologies in Table 1, it is evident that the dimensions of commercial encoders are restricted to specific lengths also requiring specific sensor components to match the length. To overcome the size limitations of encoders, this letter utilizes the characteristics of path planning using DB encoding nodes. By employing the DB sequence theorem discussed in the previous section, multiple sets of different permutations of cyclic paths can be generated for specific values of k and n. This approach satisfies the requirements for linear or circular magnetic scales of arbitrary sizes. Here, the mentioned cyclic paths represent the dimensions of the target magnetic scale or code disc. In the context of coding applications, linear scales only need to consider the path planning between the starting and ending vertices. In contrast, for circular code disc, attention must be given to the issue of repeated passage through vertices. This means that at the junction of the starting and ending vertices of the code disc, the encoding of the subarrays may overlap with repetitions on the path. Hence, this is also the reason why the proposed sequence search specifically focuses on finding sequences that adhere to the conditions of an Eulerian circuit, resolving the aforementioned issues.

Using B(25) sequence as an example for absolute encoding for a linear magnetic scale, the following will provide a comprehensive explanation of the challenges encountered when generating sets of DB sequence. For ease of identification, the sequence will employ an all-zero and all-one subsequence, marked as red text in Table 2, as the initial conditions and features encoding for the recognition of the starting and ending points of the scale. The cycle path length (ST) is first calculated as the number of bits in the cyclic encoding. After multiple recursive iterations, the sequence union of B(25) is reached, as illustrated in Table 2. With both starting and ending feature subsequences occupying 5 bits each, the ST varies from a minimum of 10 to a maximum determined by $k^n = 32$. However, certain sequences face challenges, such as vertex mismatch (ST:11, 31) or issues with repetitive encoding (ST:27, 29, 30), preventing alignment with the desired dimensions. To address these challenges, we propose a method for compensating the omissions in B(25) ST by modulating the k and n parameters, thereby introducing variability in encoder dimensions. As evident in Table 2, the extension of subarray length n can fulfill the missing lengths within the range; or alternatively, increasing the number of symbols {0, 1, 2} can satisfy the requirements. Moreover, the proposed method benefits from greater vertex variations, allowing for the quicker identification of nonrepetitive corresponding dimensions. On the other hand, the actual stroke of the encoder can be determined by multiplying the length of the cyclic encoding path by the pitch of a bit. The distance corresponding to the width between magnetic poles is also known as the pole pitch.

Table 1. Comparison of Commercial Magnetic Encoding Technologies

Brand	RLS	Magnescale	Bogen	**NTHU**
Series	LA11	SR67	LMS3	
Type	Exposed	Sealed	Exposed	Exposed
Track num.	2	2	3	2
Code	PRBS	M	Nonius	DB
Length Max. (m)	16.3	3.64	1.5	Arbitrary
Pole pitch (mm)	2	<0.1	1.28	1

Table 2. Set of B(k, n) Sequence

ST	B(2, 5) Sequence	ST	B(2, 5) Sequence
10	0000011111	22	0000010100100011011111
11	NAN	23	0000011001000101001111
12	0000010111111	24	0000010101101001000111111
13	0000011011111	25	0000010101110100100011111
14	0000011101111	26	0000011101010001001101111
15	0000010100111111	27	0000010100011100100110111111
16	0000011001011111	28	0000010010110101000111011111
17	0000010101101111	29	0000011010111001010001001111
18	0000011001110111111	30	0000010011011000111001010111111
19	0000010100011011111	31	NAN
20	0000010010101101111	32	0000010101100010001101110100111111
21	0000010101110100111111	**Compensate: B(2, 6) Sequence**	
Compensate: B(2, 4) Sequence		27	0000000110111010111100111111
11	0000101111	29	0000000110100100111101101111111
Compensate: B(3, 3) Sequence		30	0000001110111101010111001111111
11	0002021211111	31	0000001000011011101101000111111

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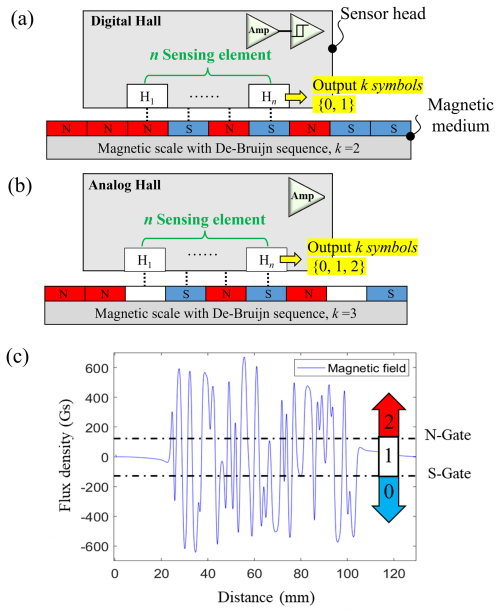


Fig. 2. Schematic illustrations of a sensor head and a magnetic medium with a DB code patter showing (a) use of a digital Hall when symbol $k = 2$ and (b) when symbol $k = 3$, and the analog Hall needs to be used to discern multiple segments to match the position states in (c).

C. Encoding Recognition Mechanism

The choice of sensors in a measurement system depends on the ability to clearly recognize the changes in magnetic poles on the magnetic scale. Normally used magnetoresistive and Hall-effect sensing elements can both effectively convert magnetic field changes into voltage signals. However, considering the impact of sensing area and circuit design on the resolution of absolute positioning, this study employs Hall-effect sensors, known for their higher fault tolerance, arranged in an array configuration. By supplying an external voltage and utilizing a differential amplifier to eliminate common-mode voltage, in conjunction with a Schmitt trigger to set voltage thresholds, the analog voltage signal can be categorized. Based on the output characteristics of the Hall sensor mentioned above, it can be applied to DB sequences for symbol conversion recognition.

In the application of magnetic encoders, traditional binary encoding utilizes digital Hall-effect sensors (H_1, H_2,…,H_n). As shown in Fig. 2(a), when the magnetic field exceeds the upper threshold, it is identified as "1"; when the magnetic field falls below the lower threshold, it is identified as "0." These digital values correspond to the magnetic north and south poles on the magnetic medium. To meet the sequence compensation strategy proposed in Table 2, this letter introduces the use of analog Hall-effect sensors to represent three sets of symbol changes {0, 1, 2}, corresponding to magnetic north, magnetic south, and no-polarity as encoding feature points. As depicted in Fig. 2(b), the term "no-polarity" refers to the option of not magnetizing the medium, in addition to inducing magnetic north and south poles. The final voltage signal will undergo system postprocessing to divide the waveform amplitude into multiple segments, as illustrated in Fig. 2(c), to match the position states of each sensing element.

III. FEASIBILITY OF MEASUREMENT SYSTEM

A. Sensor Prototype Realization

According to the sensor element schematic shown in Fig. 2, smaller sizes help improve absolute position resolution and practicality tightened housing of the sensor module. However, commercially available

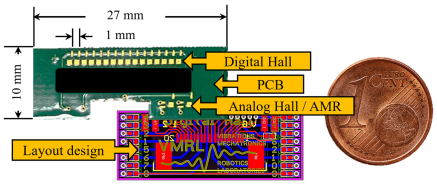


Fig. 3. Demonstration of the raw dies arranged in a 1-D array on an in-house designed of custom-made PCB with 1 mm spaced pads.

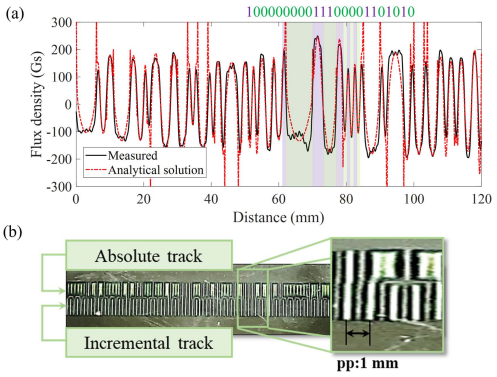


Fig. 4. (a) Comparison of measured and predicted magnetic field distribution. (b) Photography showing the magnetized result on a magnetic medium, exhibiting a dual-track pattern.

Hall-effect sensors typically have larger packaging, making integration into smaller enclosures challenging. Therefore, in the custom-printed circuit board depicted in Fig. 3, raw dies with a size of 0.56 mm are arranged in a single-dimensional array. This design allows for the adjustment of the number of Hall-effect sensing elements based on the cyclic encoding requirements of different subarray lengths. The figure also indicates that the sensor elements, constrained by the pads, are arranged at equal intervals of 1 mm to ensure the synchrony of subsequence shifts. The right-hand side of the circuit board handles pull-up resistors, filtering, and wiring connections. In addition, to enhance the overall system's absolute position resolution, the encoding adopts a dual-track design combined with the subdividable characteristics of incremental encoding periodic signals. This compensates for the minor positional variations between magnetic poles in the absolute track. In other words, 1 mm resolution of the absolute track represents the integer part of the position signal, while the fractional part is divided by the incremental track.

B. Magnetic Scale Field Estimation

Unlike the predictable periodic signals in incremental encoders, the magnetic poles in practical DB absolute encoding are arranged randomly, resulting in a more complex magnetic field distribution. This complexity is particularly important for Hall-effect sensors that output signals triggered by threshold values. Given this, we used MATLAB to construct a magnetic field distribution analysis model similar to the actual absolute encoding magnetic field. The results shown in Fig. 4(a) demonstrate that this mathematical model effectively reflects the real magnetic pole variations of absolute encoding. After applying the system's threshold settings, the custom Hall-effect sensors can accurately recognize magnetic pole states and output corresponding digital signals, validating the feasibility of the quasi-absolute encoder even with complex DB sequences.

To ensure sufficient external magnetic field strength to trigger the Hall-effect sensor, a unipolar magnetization method is employed to stably generate magnetic pole waveforms on the magnetic medium. In addition, to enhance the accuracy of the encoding magnetization, an automated absolute magnetization system was developed. Fig. 4(b)

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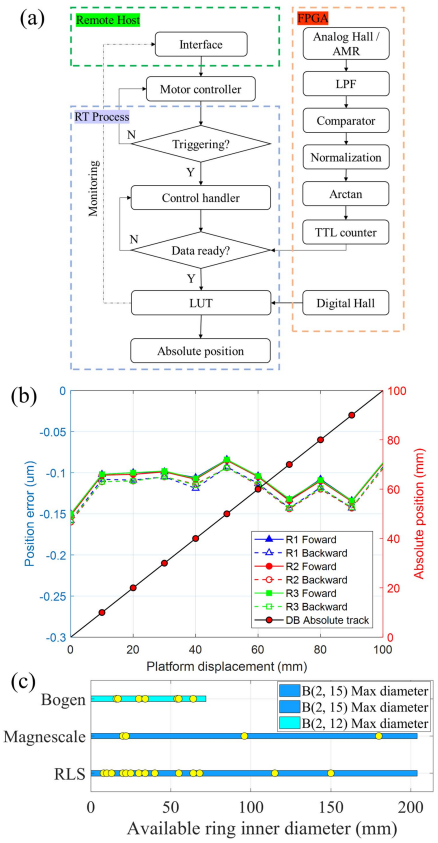


Fig. 5. (a) System's signal flow chart diagram. (b) Verification of the dual-track quasi-absolute magnetic encoder. (c) Illustration of available diameter of commercially axis magnetic encoders being matched, replaced, and even exceeded by modulating the B(k, n) values.

shows the dual-track magnetic pattern on the magnetic film card, representing the magnetization result after introducing the DB encoding sequence into the system. The upper track represents the DB absolute encoding, while the lower track represents the incremental encoding.

IV. DUAL TRACK VALIDATION

Summarizing the aforementioned DB encoding characteristics, the absolute track of this dual-track position measurement system adopts the B(2, 8) sequence. The absolute position's integer part is verified by digital Hall-effect sensors, while the fractional part is resolved by the analog Hall sensors on the circuit board in Fig. 3, which parse the periodic changes in magnetic poles in the incremental encoding. Alternatively, anisotropic magnetoresistance (AMR) sensors can be used for better stability, although this increases the cost of the sensor module. In developing the system firmware, precise and rapid capture of target position signals is essential. Therefore, the system employs a design based on field-programmable gate arrays (FPGA), utilizing programmable logic resources and configurable interconnects, allowing the system can load multiple signals into memory in as little as 8 ms. The final subsequence encoding corresponds to the absolute position output in the built-in lookup table, while subsequent positional changes are calculated using the relative movements from the incremental track. The signal flow diagram of the entire system's transmission and computation process is shown in Fig. 5(a).

To validate system performance, a linear platform with an effective stroke of 100 mm was used. By driving the linear platform to move the magnetic scale, relative motion between the sensor module and the magnetic scale was induced. A high-resolution external optical encoder positioning system with 0.1 μs accuracy was employed as a benchmark, and the positions identified by the absolute position measurement system were compared to it. As shown in the position error versus platform displacement plot in Fig. 5(b), the system using the proposed DB encoding demonstrated excellent bidirectional repeatability and accuracy across three trials (R1, R2, and R3), with precision values of 0.014 and 0.072 mm for forward and backward measurements, respectively. In addition, in terms of absolute positioning, the system aligned precisely with the integer part of the optical scale, indicating that the dual-track DB magnetic cyclic encoding and sensor module used in this study are feasible and can be extended to the quasi-absolute encoders of arbitrary sizes. Fig. 5(c) shows the available diameter (circle marker) of commercial axis encoders can be matched using different B(k, n) combinations, and the corresponding stroke can be filtered from this series.

V. CONCLUSION

This letter demonstrates the potential of DB sequences to accommodate quasi-absolute encoders of arbitrary sizes, effectively overcoming the limitations of commercial encoder specifications. By utilizing custom single-array Hall-effect sensors and a dual-track magnetic scale, we have validated the feasibility of high-precision quasi-absolute encoders. Our findings highlight the high adaptability of DB sequences to linear scales and code discs, which can significantly reduce manufacturing costs. Furthermore, the ability to increase symbol variations allows for breaking away from the traditional binary framework, underscoring the superiority and confidentiality of magnetic encoders in sequence arrangement compared to optical encoders. While the quasi-absolute encoder does not achieve the resolution of optical encoders, it provides sufficient resolution for most applications.

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